

ORIGINAL

ThreeBond

SAFETY DATA SHEET

COPY

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Version 2

Section 1: PRODUCT AND COMPANY IDENTIFICATION

Product identifier**Product name** ThreeBond 1521**Recommended use of the chemical and restrictions on use****Recommended use** Adhesive, Sealant**Details of the supplier of the safety data sheet****Manufacturer**

ThreeBond Fine Chemical Co., Ltd.

Department in charge & AddressProduction Engineering Division
1-1 Oyama-cho, Midori-ku
Sagamihara-shi, Kanagawa, Japan**Emergency telephone number**

+81-42-774-1333

Section 2: HAZARDS IDENTIFICATION

Classification of the substance or mixture

Flammable liquids	Category 2
Acute toxicity - Inhalation (Vapors)	Category 4
Skin corrosion/irritation	Category 2
Serious eye damage/eye irritation	Category 2
Reproductive Toxicity	Category 1A
Specific target organ toxicity (single exposure)	Category 1
Category 1 Central nervous system	
Category 3 Respiratory irritation, Narcotic effects.	
Specific target organ toxicity (repeated exposure)	Category 1
Category 1 Central nervous system, kidneys	
Category 2 nervous system.	
Aspiration toxicity	Category 1
Acute aquatic toxicity	Category 2
Chronic aquatic toxicity	Category 3

Label elements

Signal word

Danger

Hazard statements

H225 - Highly flammable liquid and vapor

H304 - May be fatal if swallowed and enters airways
H315 - Causes skin irritation
H319 - Causes serious eye irritation
H332 - Harmful if inhaled
H360 - May damage fertility or the unborn child
H370 - Causes damage to organs
H372 - Causes damage to organs through prolonged or repeated exposure
H401 - Toxic to aquatic life
H412 - Harmful to aquatic life with long lasting effects
H335 - May cause respiratory irritation
H336 - May cause drowsiness or dizziness

Causes damage to the following organs: Central nervous system.

Causes damage to the following organs through prolonged or repeated exposure: Central nervous system, kidneys.

May cause damage to the following organs through prolonged or repeated exposure: nervous system.

Precautionary Statements - Prevention

- Obtain special instructions before use
- Do not handle until all safety precautions have been read and understood
- Use personal protective equipment as required
- Use only outdoors or in a well-ventilated area
- Wash face, hands and any exposed skin thoroughly after handling
- Do not breathe dust/fume/gas/mist/vapors/spray
- Do not eat, drink or smoke when using this product
- Avoid release to the environment.
- Keep away from heat/sparks/open flames/hot surfaces. — No smoking
- Keep container tightly closed
- Ground/bond container and receiving equipment
- Use explosion-proof electrical/ventilating/lighting/equipment
- Use only non-sparking tools
- Take precautionary measures against static discharge

Precautionary Statements - Response

- For first aid procedure, refer to this SDS.
- IF exposed: Call a POISON CENTER or doctor/physician
- IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
- If eye irritation persists: Get medical advice/attention
- If skin irritation occurs: Get medical advice/attention
- IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/shower
- Wash contaminated clothing before reuse
- IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing
- Call a POISON CENTER or doctor/physician if you feel unwell
- IF SWALLOWED: Immediately call a POISON CENTER or doctor/physician
- Do NOT induce vomiting
- In case of fire: Use CO2, dry chemical, or foam for extinction

Precautionary Statements - Storage

- Store locked up
- Store in a well-ventilated place. Keep cool

Precautionary Statements - Disposal

- Dispose of contents/container to an approved waste disposal plant

Other hazards

- May be harmful if swallowed

Section 3: COMPOSITION/INFORMATION ON INGREDIENTS**Single substance or mixture**

Mixture

Effective June 1, 2016, regarding Japan's Industrial Safety and Health Law's "Notifiable Dangerous and Harmful", target substances will be subjected to risk assessment in accordance with Japan's Industrial Safety and Health Law's "Harmful Substances Whose Names Are to be Indicated on the Label."

Chemical name	Weight-%	ENCS	ISHL No.	CAS No.
Toluene	39	(3)-2	-	108-88-3
Hexane	22	(2)-6	-	110-54-3
Ethyl acetate	10-20	(2)-726	-	141-78-6
Chloroprene Synthetic rubber, Phenol resin	20-30	-	-	-

Pollution Release and Transfer Registry

Class	Chemical Name in Regulation	(Metal Name)	Ordinance Number
First Class Designated Chemical Substances (Law Art. 2-2, Enforcement Order Art. 1 Attached Table No.1)	Toluene	-	300
First Class Designated Chemical Substances (Law Art. 2-2, Enforcement Order Art. 1 Attached Table No.1)	n-Hexane	-	392

Industrial Safety and Health Law

Law Name	Chemical Name in Regulation	Ordinance Number
Harmful Substances Whose Names Are to be Indicated on the Label (Law Art.57, Para. 1, Enforcement Order Art.18)	Ethyl acetate	9-9
Harmful Substances Whose Names Are to be Indicated on the Label (Law Art.57, Para. 1, Enforcement Order Art.18)	Toluene	23
Harmful Substances Whose Names Are to be Indicated on the Label (Law Art.57, Para. 1, Enforcement Order Art.18)	n-Hexane	27
Notifiable Substances (Law Art.57-2, Enforcement Order Art.18-2 Attached Table No.9)	Ethyl acetate	177
Notifiable Substances (Law Art.57-2, Enforcement Order Art.18-2 Attached Table No.9)	Toluene	407
Notifiable Substances (Law Art.57-2, Enforcement Order Art.18-2 Attached Table No.9)	Hexane	520

Act on the Evaluation of Chemical Substances and Regulation of Their Manufacture, etc

Law Name	Chemical Name in Regulation	Ordinance Number
Priority Assessment Chemical Substances (Law Article 2, Para.5)	n-hexane	3
Priority Assessment Chemical Substances (Law Article 2, Para.5)	Toluene	46

Section 4: FIRST AID MEASURES**INHALATION**

Move victim to fresh air If breathing is irregular or stopped, administer artificial respiration
Administer oxygen if breathing is difficult

Skin contact

Wash skin with soap and water

Eye contact

In case of contact with substance, immediately flush skin or eyes with running water for at least 20 minutes

INGESTION

Rinse mouth. Get medical attention.

Self-protection of the first aider Ensure that medical personnel are aware of the material(s) involved and take precautions to protect themselves.

Note to physicians Keep victim warm and quiet.

Section 5: FIRE FIGHTING MEASURES

Flammable properties HIGHLY FLAMMABLE: Will be easily ignited by heat, sparks or flames. Containers may explode when heated. Many liquids are lighter than water.

Suitable extinguishing media Dry chemical, CO2, water spray or regular foam Water spray, fog or regular foam Use water spray or fog; do not use straight streams

Unsuitable extinguishing media CAUTION: All these products have a very low flash point. Use of water spray when fighting fire may be inefficient.

Specific hazards arising from the chemical Vapors may form explosive mixtures with air Vapors may travel to source of ignition and flash back Most vapors are heavier than air. They will spread along ground and collect in low or confined areas (sewers, basements, tanks) Vapor explosion hazard indoors, outdoors or in sewers Those substances designated with a "P" may polymerize explosively when heated or involved in a fire Runoff to sewer may create fire or explosion hazard Substance may be transported hot

Special extinguishing media Wear protection gear and extinguish from windward.

Section 6: ACCIDENTAL RELEASE MEASURES

Personal precautions ELIMINATE all ignition sources (no smoking, flares, sparks or flames in immediate area) All equipment used when handling the product must be grounded Do not touch or walk through spilled material Stop leak if you can do it without risk

Environmental precautions Prevent entry into waterways, sewers, basements or confined areas

Methods for containment A vapor suppressing foam may be used to reduce vapors Absorb or cover with dry earth, sand or other non-combustible material and transfer to containers

Methods for cleaning up Use clean non-sparking tools to collect absorbed material. Dike far ahead of liquid spill for later disposal.

Prevention of secondary hazards Keep ignition source away from spill.

Section 7: HANDLING AND STORAGE

Handling

Precautions for safe handling

Advice on safe handling Take equipment measures listed in Section 8. Wear protection gear.

Local and general ventilation Take equipment measures listed in Section 8. Wear protection gear.

Storage

Storage conditions Close lid. Avoid direct sun light and ignition source. Keep appropriate temperature.

Material of vessels and packaging Keep this product in original container. Do not put it back in the container.

Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Exposure guidelines

Chemical name	Japan	ISHL Working Environmental Evaluation Standards - Administrative Control Levels	ACGIH TLV
Toluene	TWA: 50 ppm TWA: 188 mg/m ³ Skin ISHL/ACL: 20 ppm	ISHL/ACL: 20 ppm	TWA: 20 ppm
Hexane	TWA: 40 ppm TWA: 140 mg/m ³ Skin ISHL/ACL: 40 ppm	ISHL/ACL: 40 ppm	TWA: 50 ppm Skin
Ethyl acetate	TWA: 200 ppm TWA: 720 mg/m ³ ISHL/ACL: 200 ppm	ISHL/ACL: 200 ppm	TWA: 400 ppm

Engineering controls

Install local ventilation or seal source of substances. Install safety shower, hand wash, and eye wash station. Clearly indicate the location.

Personal protective equipment

- ☐ Respiratory protection In case of inadequate ventilation wear respiratory protection
- ☐ Hand protection Wear appropriate protection glove (Made from non-permeable material such as polyethylene, rubber)
- ☐ Eye/face protection Wear safety glasses with side shields (or goggles)
- ☐ Skin and body protection Wear protection apron, protection boots. Wear long sleeve cloth.

Other information

Wash hands thoroughly after handling. When using do not eat, drink or smoke.

Section 9: PHYSICAL AND CHEMICAL PROPERTIES

Physical state Liquid
Odor Solvent odor
Color Brown

<u>Property</u>	<u>Values</u>	<u>Remarks</u>
pH	No data available	
Melting point/freezing point	No data available	
Boiling point / boiling range	67 °C or above	
Flash point	-20 °C	
Evaporation rate	No data available	
Flammability (solid, gas)		
Flammability limit in air		
Upper flammability limit:	No data available	
Lower flammability limit:	No data available	
Specific gravity	0.88	
Water solubility	Slightly soluble	
Autoignition temperature	No data available	
Decomposition temperature	No data available	
Dynamic viscosity	2.8 Pa·s	

Section 10: STABILITY AND REACTIVITY

Stability Stable under normal conditions.

Possibility of hazardous reactions React with strong acid. Could cause fire.

Conditions to avoid Heat

Incompatible materials Strong oxidizing agents.

Hazardous decomposition products May generate harmful gas by incineration

Section 11: TOXICOLOGICAL INFORMATION

Acute toxicity

The following values are calculated based on chapter 3.1 of the GHS document

Inhalation LC50 No data available as this product.

Numerical measures of toxicity - Component Information

Chemical name	Oral LD50	Dermal LD50	Inhalation LC50
Toluene	= 2600 mg/kg (Rat)	= 12000 mg/kg (Rabbit)	= 12.5 mg/L (Rat) 4 h
Hexane	= 25 g/kg (Rat)	= 3000 mg/kg (Rabbit)	= 48000 ppm (Rat) 4 h
Ethyl acetate	= 5620 mg/kg (Rat)	> 18000 mg/kg (Rabbit) > 20 mL/kg (Rabbit)	-

Delayed and immediate effects as well as chronic effects from short and long-term exposure

Skin corrosion/irritation No data available as this product.

Serious eye damage/eye irritation No data available as this product.

Sensitization No data available as this product.

Germ cell mutagenicity No data available as this product.

Carcinogenicity The table below indicates whether each agency has listed any ingredient as a carcinogen

Chemical name	Japan	IARC
Toluene		Group 3

*IARC (International Agency for Research on Cancer)
Not classifiable as a human carcinogen*

Reproductive toxicity No data available as this product.

STOT - single exposure No data available as this product.

STOT - repeated exposure No data available as this product.

Target organ effects Central nervous system, Eyes, kidney, liver, Peripheral Nervous System (PNS), Respiratory system, Skin.

Aspiration hazard No data available as this product.

Section 12: ECOLOGICAL INFORMATION

Ecotoxicity

Acute aquatic hazard No data available as this product.

Chronic aquatic hazard No data available as this product.

Ecotoxicity Toxic to aquatic life Harmful to aquatic life with long lasting effects

Chemical name	Algae/aquatic plants	Fish	Crustacea
Toluene	433: 96 h <i>Pseudokirchneriella subcapitata</i> mg/L EC50 12.5: 72 h <i>Pseudokirchneriella subcapitata</i> mg/L EC50 static	15.22 - 19.05: 96 h <i>Pimephales promelas</i> mg/L LC50 flow-through 12.6: 96 h <i>Pimephales promelas</i> mg/L LC50 static 5.89 - 7.81: 96 h <i>Oncorhynchus mykiss</i> mg/L LC50 flow-through 5.8: 96 h <i>Oncorhynchus mykiss</i> mg/L LC50 semi-static 11.0 - 15.0: 96 h <i>Lepomis macrochirus</i> mg/L LC50 static 54: 96 h <i>Oryzias latipes</i> mg/L LC50 static 14.1 - 17.16: 96 h <i>Oncorhynchus mykiss</i> mg/L LC50 static 28.2: 96 h <i>Poecilia reticulata</i> mg/L LC50 semi-static 50.87 - 70.34: 96 h <i>Poecilia reticulata</i> mg/L LC50 static	11.5: 48 h <i>Daphnia magna</i> mg/L EC50 5.46 - 9.83: 48 h <i>Daphnia magna</i> mg/L EC50 Static
Hexane	-	2.1 - 2.98: 96 h <i>Pimephales promelas</i> mg/L LC50 flow-through	>1000: 24 h <i>Daphnia magna</i> mg/L EC50
Ethyl acetate	3300: 48 h <i>Desmodesmus subspicatus</i> mg/L EC50	352 - 500: 96 h <i>Oncorhynchus mykiss</i> mg/L LC50 semi-static 220 - 250: 96 h <i>Pimephales promelas</i> mg/L LC50 flow-through 484: 96 h <i>Oncorhynchus mykiss</i> mg/L LC50 flow-through	560: 48 h <i>Daphnia magna</i> mg/L EC50 Static

Persistence and degradability No data available as this product.

Bioaccumulation No data available as this product.

Mobility in soil No data available as this product.

Endocrine disruptor information No data available as this product.

Section 13: DISPOSAL CONSIDERATIONS

Waste from residues / unused products Dispose of in accordance with national, state and local regulations. Consult industrial waste management companies for waste. Do not release this product to natural environment nor reclaim.

Contaminated packaging Dispose containers as same as residual of this product.

Section 14: TRANSPORT INFORMATION

IMDG

UN/ID No.	UN1133
Proper shipping name	Adhesives
Hazard class	3
Packing group	III
EmS-No	F-E, S-D

ICAO/IATA (air)

UN/ID No.	UN1133
Proper shipping name	Adhesives
Hazard class	3
Packing group	III

ADR

UN/ID No.	UN1133
Proper shipping name	Adhesives
Hazard class	3
Packing group	III
ERG code	3L

Japanese regulations

UN Number	UN1133
Proper shipping name	Adhesives
Hazard class	3
Packing group	III
Marine Transportation Safety Act	Flammable Liquids (Ministry of Transportation Ordinance Regarding Transport by Ship and Storage, Ordinance Art.3, Attached Table 1)
Civil Aeronautics Act	Flammable Liquids (MITL Notification for Air Transportation of Explosives etc., Ordinance Art.194, Attached Table 1)

Section 15: REGULATORY INFORMATION

Effective June 1, 2016, regarding Japan's Industrial Safety and Health Law's "Notifiable Dangerous and Harmful", target substances will be subjected to risk assessment in accordance with Japan's Industrial Safety and Health Law's "Harmful Substances Whose Names Are to be Indicated on the Label."

Fire protection law criteria Group 4 - Petroleum - 1st Class(not Water solubility)

Act on the Evaluation of Chemical Substances and Regulation of Their Manufacture, etc Priority Assessment Chemical Substances (Law Article 2, Para.5)

Industrial Safety and Health Law Harmful Substances Whose Names Are to be Indicated on the Label (Law Art.57, Para.1, Enforcement Order Art.18)
Notifiable Substances (Law Art.57-2, Enforcement Order Art.18-2 Attached Table No.9)

Act on Confirmation, etc. of Release Amounts of Specific Chemical Substances in the Environment and Promotion of Improvements to the Management Thereof First Class Designated Chemical Substances (Law Art. 2-2, Enforcement Order Art. 1 Attached Table No.1)

Section 16: OTHER INFORMATION

Issue date 29-Aug-2014

Other information Please contact to local sales offices for further information.

Disclaimer

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